Airline Booking ERD

Unary Relationship

- **CUSTOMER.referred_by**
    - referred_by is a foreign key on the CUSTOMER table that points back to customer_id on the same table.
    - A customer can refer other customers to the platform.
    - One customer can refer many others (one-to-many).
    - NULL means the customer was not referred by anyone.
    - Shown in the ERD as a loop line from CUSTOMER back to itself.

Ternary (3-Way) Relationship

- **BOOKING_FLIGHT**
    - BOOKING_FLIGHT is a ternary associative entity connecting three tables simultaneously: BOOKING, FLIGHT, and PASSENGER.
    - Each row means a specific passenger is on a specific flight under a specific booking, one ticket.
    - You cannot split this into binary pairs without losing information.
    - It carries its own attribute: ticket_price.
    - SEAT_ASSIGNMENT and LUGGAGE both connect to BOOKING_FLIGHT, not directly to FLIGHT or PASSENGER, because a seat and luggage belong to a specific passenger on a specific flight under a specific booking.

Disjoint Subtype Hierarchy

- **Supertype: PASSENGER**
    - PASSENGER is the supertype. Every traveler on a plane is a passenger.
    - The passenger_type column is constrained to exactly three values: adult, minor, or employee.
    - customer_id is nullable because employees are not customers.
- **Subtype 1: ADULT_PASSENGER**
    - Represents paying adult travelers.
    - Adds: date_of_birth and frequent_flyer_number.
    - passenger_id is both the PK of this table and a FK back to PASSENGER.
- **Subtype 2: MINOR_PASSENGER**
    - Represents child travelers.
    - Adds: date_of_birth and guardian_passenger_id.

- guardian_passenger_id is a FK to ADULT_PASSENGER, a minor must have an adult guardian.

- **Subtype 3: EMPLOYEE_PASSENGER**
  - Represents crew members: pilots, co-pilots, flight attendants.
  - Adds: employee_number, job_title, hire_date, and airline_id.
  - airline_id is a FK to AIRLINE because crew are employed by a specific airline.
  - customer_id is NULL for all employees, they are not customers.
- **Disjoint**
  - A passenger is exactly one subtype, never two at once. This is enforced by the CHECK constraint on passenger_type.
  - In a formal ER diagram this is shown with a circle labeled 'd' and an arc connecting the three subtypes to the supertype.

Binary Relationships

- **CUSTOMER -> BOOKING**
  - One customer can make many bookings. Each booking belongs to one customer.
- **BOOKING -> PAYMENT**
  - One booking has one payment. Each payment belongs to one booking.
- **BOOKING -> BOOKING_FLIGHT**
  - One booking can have many booking_flight rows (multiple flights or passengers per booking).
- **FLIGHT -> BOOKING_FLIGHT**
  - One flight appears in many booking_flight rows across different bookings.
- **PASSENGER -> BOOKING_FLIGHT**
  - One passenger can appear in many booking_flight rows across different bookings and flights.
- **PASSENGER -> CUSTOMER**
  - A passenger optionally links to a customer. Paying passengers have a customer_id. Employees have NULL.
- **BOOKING_FLIGHT -> SEAT_ASSIGNMENT**
  - One booking_flight can have many seat assignments. Each seat belongs to one booking_flight.
- **BOOKING_FLIGHT -> LUGGAGE**

- One booking_flight can have many luggage add-ons. Each luggage row belongs to one booking_flight.

- **AIRLINE -> FLIGHT**
  - One airline operates many flights. Each flight belongs to one airline.
- **AIRLINE -> EMPLOYEE_PASSENGER**
  - One airline employs many crew members. Each employee belongs to one airline.
- **ROUTE -> FLIGHT**
  - One route has many scheduled flights. Each flight runs on one route.
- **AIRPORT -> ROUTE (twice)**
  - AIRPORT participates twice in ROUTE, once as origin_airport_id and once as destination_airport_id.
  - This is why there are two lines from AIRPORT to ROUTE in the diagram.
  - One airport can be the origin of many routes, and separately the destination of many routes.
- **MINOR_PASSENGER -> ADULT_PASSENGER**
  - A minor must have a guardian who is an adult passenger.
  - guardian_passenger_id on MINOR_PASSENGER is a FK to ADULT_PASSENGER.

Associative Entities

- **BOOKING_FLIGHT**
  - Resolves the many-to-many between BOOKING, FLIGHT, and PASSENGER (ternary).
  - Also acts as the anchor point for SEAT_ASSIGNMENT and LUGGAGE.
  - Carries its own attribute: ticket_price.
- **ROUTE**
  - Resolves the many-to-many between origin and destination airports.
  - Carries its own attribute: distance_miles.